Advaith Appajodu

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Education

University of California, Berkeley Berkeley, CA
B.A. in Applied Mathematics and Computer Science *2028*

- **Relevant Coursework**, Data Structures, Structure and Interpretation of Computer Programs, Linear Algebra and Differential Equations, Discrete Mathematics, Introduction to Quantitative Finance

Experience

Berkeley AI Research (BAIR) Lab, Jiao Group Oct. 2025 – Present
Student Researcher, Advisor Prof. Jiantao Jiao *Berkeley, CA*

- Collaborated on the design and delivery of a benchmark measuring whether frontier LLMs can repair their own mid-procedure reasoning errors, by building a Python pipeline that generates procedural tasks up to 150 steps deep, injects one of 6 formal edit types, and scores the repair against a deterministically resolved target build.
- Engineered the clean-room physics solver behind the scoring, by implementing grid occupancy, a placement DSL with parse and execute stages, collision detection, stud level connectivity, support based stability, and per-prefix buildability checks under unit test.
- Built the evaluation harness and experimental design behind the project's first multi-model results, by tracking reference and physical dependency graphs, reducing under-correction and over-correction to a single localization F1, and stratifying 90 runs by edit type and task depth.

MoolAI 2026 – Present
Forward Deployed Engineer Intern *San Ramon, CA*

- Built a production LLM email agent for an enterprise client that classifies inbound Gmail into 5 intent classes and drafts replies in the user's own writing voice, by pairing scoped LLM calls with a deterministic event driven pipeline built on idempotent processing and dead letter recovery.
- Improved reply voice fidelity continually while minimizing stored personal data, by diffing user edits against held drafts, folding sent mail into a per user tone corpus, and persisting only text the user authored.
- Shipped the agent with 229 passing tests, strict static typing, and a structural guarantee it can never send mail autonomously, by removing every send code path and enforcing the invariant with an automated CI check.

Projects

Budgeted Cross-Lingual Reasoning | *Independent Research Project, Python, LLM Evaluation* 2026

- Ran a controlled study spanning 900 budgeted generations across 4 open-weight LLMs and 5 reasoning languages on BIG-Bench Extra Hard, by designing a full factorial evaluation with language compliance as a first-class covariate.
- Shipped a reproducible evaluation harness covered by 56 unit tests, by implementing compliance gating, a script-agnostic degenerate-loop detector, and a normalized cross-provider response schema.
- Overturned a statistically significant but false model ranking, by isolating a serving-tier confounder, recovering 62% of truncated runs through a paired synchronous rerun, and correcting the reported results.

trAId | *Python, FastMCP, Redis, Browserbase, React, TypeScript* 2026

- Won the 2026 CalHacks AI Hackathon sponsor track.
- Cut item pricing and listing from hours to seconds, by building an autonomous resale agent that identifies, prices, and lists an item from a single photo over text through Poke.
- Delivered real-time end-to-end marketplace automation, by engineering a Python and FastMCP server backed by Redis and Browserbase with a React and TypeScript dashboard streaming agent events over SSE.

High-Performance Deep Learning Systems | *PyTorch, CUDA, CNNs, Transformers* 2025

- Reached 82% test accuracy on CIFAR-10 and 76 BLEU on English to Telugu translation, by building and training CNNs and Transformers from scratch with multi-head attention, regularization, and optimizer tuning.

Technical Skills

Languages: Python, C++, Java, Go, SQL, JavaScript/TypeScript
ML/AI: PyTorch, Transformers, LoRA, Hugging Face, LLM Evaluation, Agents, MCP, Model Evaluation, Training Pipelines
Systems: Distributed Training, Data Pipelines, GPU Workloads, Benchmarking, Observability
Web: React, Next.js, REST APIs, Mapbox, SQLite
Tools: Docker, Git, Weights & Biases, Browserbase, Redis, n8n